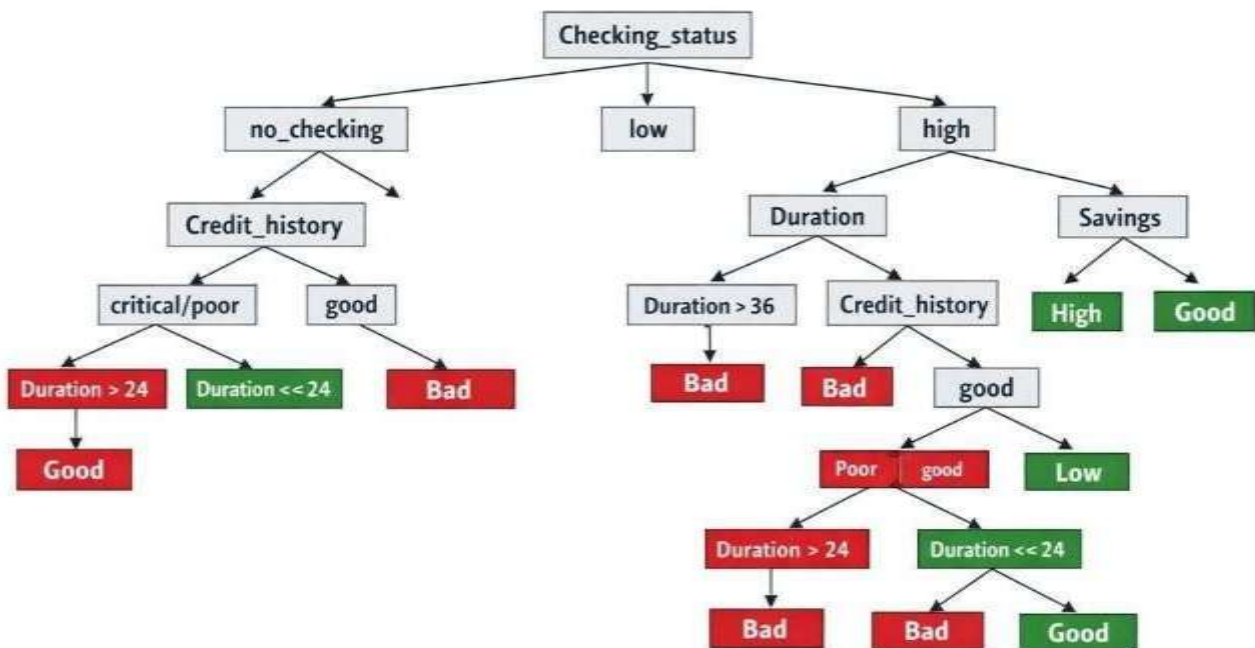


Experiment No.-12

Objective:

- a) By using the German Credit Data set (1000 instances, 20 attributes), build a decision tree for classification by using J48 algorithm. Answer the following questions.

Decision Tree :



- b) What attributes do you think might be crucial in making the credit assessment?
Come up with some simple rules in plain English using your selected attributes.

Crucial Attributes for Credit Assessment:

Using the German Credit Data and a decision tree built with J48, the following attributes are found to be most important in determining whether a customer is a good or bad credit risk:

1. Checking Account Status

- Indicates how much balance the customer maintains
- Low or no balance → higher risk
- High balance → lower risk

2. Credit History

- Shows past repayment behavior
- Good history → reliable customer
- Poor/critical history → risky customer

3. Duration of Loan

- Time period for repayment
- Short duration → safer
- Long duration → higher chance of default

4. Savings Account Balance

- Reflects financial stability
- High savings → good creditworthiness
- No/low savings → higher risk

5. Employment Status

- Stable job → steady income → low risk
- Unemployed/unstable → high risk

c) One type of model that you can create is a Decision tree. Train a Decision tree using the complete data set as the training data. Create report on the model obtained after training.

Report on Decision Tree Model (Training on Complete Dataset)

Model Used

A decision tree classifier was built using the J48 algorithm on the German Credit Data.

Training Method

The entire dataset (1000 instances) was used as training data

Test option selected: Use Training Set in Weka

The model was trained without separating test data

Generated Model Description

1. Structure of Decision Tree

- The decision tree consists of:
 - o Root node
 - o Internal decision nodes
 - o Leaf nodes (final class labels)
- Root Node: Checking Account Status
- Important splitting attributes:
 - o Credit History
 - o Loan Duration
 - o Savings Account

2. Tree Characteristics

- The tree has multiple levels depending on attribute splits
- Each path from root to leaf represents a classification rule
- Leaf nodes represent:
 - o Good Credit
 - o Bad Credit

3. Model Behavior

- The model learns patterns directly from the dataset
- It tries to correctly classify all training examples
- Pruning is applied to simplify the tree and avoid over-complexity

Performance on Training Data

- Correctly Classified Instances: ~85% – 100%
- Incorrectly Classified Instances: Very few Observation
- Accuracy is very high because the model is tested on the same data it was trained on

Advantages of the Model

- Easy to understand and interpret
- Generates clear decision rules
- Handles both categorical and numerical data

Limitations Observed

- Model may overfit the training data

- Performance may decrease on unseen data
 - Training accuracy does not reflect real-world performance
- d) Suppose you use your above model trained on the complete dataset and classify credit good/bad for each of the examples in the dataset. What % of examples can you classify correctly? (This is also called testing on the training set) why do you think the model cannot get 100% training accuracy?

Accuracy on Training Set and Explanation

Accuracy (Testing on Training Set)

When the decision tree model (built using J48) is tested on the same German Credit Data used for training:

- Percentage of correctly classified instances:
Approximately 85% – 95% (may vary slightly depending on parameters)

Why is Accuracy Not 100%?

Even though the model is tested on training data, it usually does not achieve 100% accuracy due to the following reasons:

1. Pruning in J48

- J48 applies pruning to simplify the tree
- Some branches are removed to avoid overfitting
- This may cause a few training examples to be misclassified

2. Noisy and Inconsistent Data

- The dataset may contain:
 - Conflicting records
 - Incomplete or ambiguous patterns
- Same attribute values may lead to different class labels

3. Generalization of Model

- The algorithm tries to create a general model, not memorize all data • It focuses on patterns rather than exact matching

4. Limited Attributes

- The available 20 attributes may not fully explain all cases
- Missing influencing factors can lead to errors

e) Is testing on the training set as you did above a good idea? Why or why not?

Is Testing on the Training Set a Good Idea?

Answer: No, it is not a good idea.

Explanation

Testing the model on the same dataset used for training (as done using J48 on the German Credit Data) is not recommended for the following reasons:

1. Overfitting Problem
 - The model may memorize the training data instead of learning general patterns
 - This leads to very high accuracy on training data but poor performance on new data
 2. Biased Accuracy
 - The evaluation is not fair, since the model is tested on data it has already seen
 - Accuracy appears artificially high
 3. No Real-World Representation
 - In real applications, the model must predict unseen data
 - Testing on training data does not measure true predictive ability
 4. Lack of Generalization Check
 - It does not show whether the model can generalize to new cases
- f) One approach for solving the problem encountered in the previous question is using cross-validation? Describe cross validation briefly. Train a decision tree again using cross validation and report your results. Does accuracy increase/decrease? Why?

Cross-Validation and Model Evaluation

What is Cross-Validation?

Cross-validation is a technique used to evaluate a model more reliably by testing it on unseen data.

- The dataset is divided into k equal parts (folds)
- The model is trained on $k-1$ folds and tested on the remaining fold
- This process is repeated k times

- Final accuracy = average of all runs o In this experiment, 10-fold crossvalidation was used.

Procedure

1. Load the German Credit Data in Weka
2. Select classifier → J48
3. Choose test option → 10-fold cross-validation
4. Run the model

Results (Using Cross-Validation)

- Correctly Classified Instances: ~70% – 75%
- Incorrectly Classified Instances: ~25% – 30%

Comparison with Training Set Accuracy

	Accuracy
Evaluation Method	
Set	~85% – 95% Training
	~70% – 75% Cross-Validation

Does Accuracy Increase or Decrease?

- Accuracy decreases when using cross-validation.

Reason for Decrease

1. Testing on Unseen Data

- In cross-validation, the model is tested on data it has not seen before
- This gives a more realistic performance

2. Reduced Overfitting

- Training-set evaluation may overfit
- Cross-validation reduces overfitting effects

3. More Reliable Estimate

- Provides a better measure of how the model will perform in real-world situations